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Title of the copyright:

A Novel System for Keylogger Detection using Dendritic Cell Algorithm and Machine Learning Algorithms based on Typing Speed

Copyright Work:

The Flowchart of keylogger detection is shown in figure 1.

The input to the system can be normal user keystrokes or commands and the keylogging commands given by the keylogger.

Step 1: The dataset generated from the above inputs is given as input to the Classification Algorithm Support Vector Machine. SVM will classify the dataset into the input whose speed is greater than or equal to 60 and less than 60.

Step 2: The inputs having speed less than 60 can have keylogging commands present, hence this generated dataset of the inputs having speed less than 60 is given as input to the Classification Algorithm Naïve Bayes. Naïve Bayes will Classify the dataset into the inputs who are having same time intervals between them and inputs with no fix time interval.

Step 3: According to traffic analysis for keylogger detection the if 4 or more consecutive commands have same time interval between are keylogger commands. These keylogging commands caught in the system which assures that keylogger is present in the system. The commands which do not have same time interval between them can be safe user commands.

Step 4: Further the commands having speed less than 60 and commands having speed greater than or equal to 60 given as input to the Dendritic Cell Algorithm which generates high Antigens & Signals. This leads to successful Keylogger Detection in the system.

Step 5: If the commands do not have same time interval between them the keylogger is not sending commands with same time interval it means the commands are not periodic still this system can detect the keylogger present in the system as the data having speed greater than or equal to 60 is given as input to Dendritic Cell Algorithm which generates high Antigens & signals and leads to successful keylogger detection.

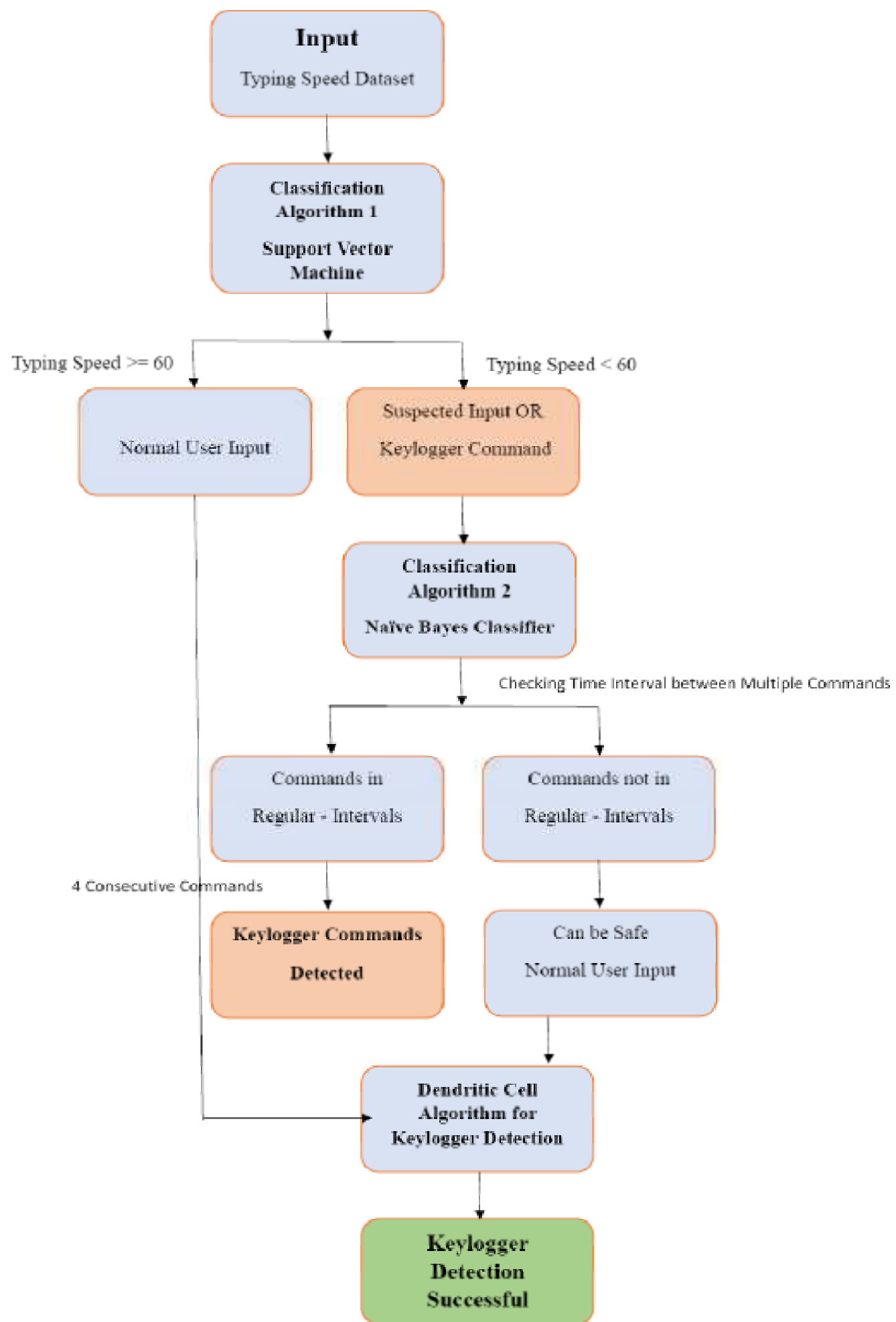


Figure 1: Flowchart of Proposed system